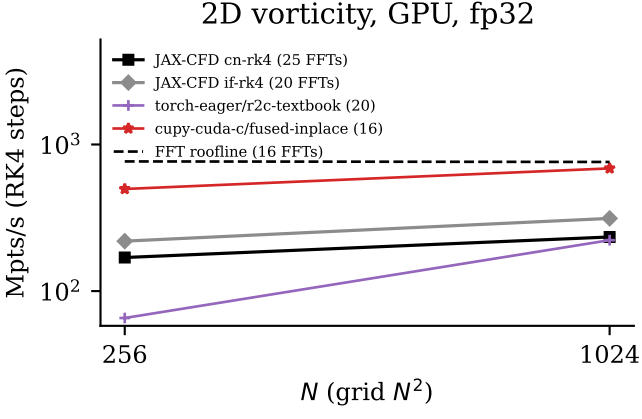
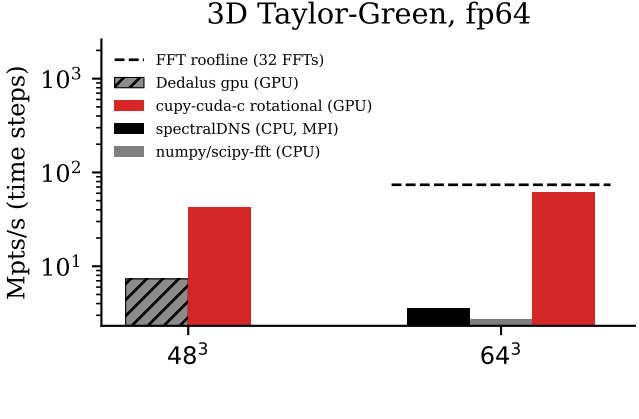

2D vorticity, GPU, fp32
JAX-CFD cn-rk4 (25 FFTs)
JAX-CFD if-rk4 (20 FFTs)
torch-eager/r2c-textbook (20)
cupy-cuda-c/fused-inplace (16)
FFT roofline (16 FFTs)
Mpts/s (RK4 steps)
$10^3$
$10^2$
256
1024
$N$ (grid $N^2$)


3D Taylor-Green, fp64
FFT roofline (32 FFTs)
Dedalus gpu (GPU)
cupy-cuda-c rotational (GPU)
spectralDNS (CPU, MPI)
numpy/scipy-fft (CPU)
Mpts/s (time steps)
$10^3$
$10^2$
$10^1$
$48^3$
$64^3$